

First-Order ODEs

Zill 2.2–2.5

|                                                         |                                                                                                                                                             |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Separable</b><br>separate, integrate                 | $\frac{dy}{dx} = g(x)h(y)$ $\int \frac{dy}{h(y)} = \int g(x) \, dx + C$                                                                                     |
| <b>Linear</b><br>integrating factor $\mu$               | $y' + P(x)y = Q(x), \quad \mu(x) = e^{\int P(x) \, dx}$ $y = \frac{1}{\mu(x)} \left[ \int \mu(x)Q(x) \, dx + C \right]$                                     |
| <b>Exact</b><br>potential fn. $f$                       | $M \, dx + N \, dy = 0 \quad \text{exact iff} \quad M_y = N_x$ $f = \int M \, dx + g(y), \quad \text{solve } g \text{ from } f_y = N$ $f(x, y) = C$         |
| <b>Non-exact</b><br>restore $M_y = N_x$                 | $\mu(x) = e^{\int \frac{M_y - N_x}{N} \, dx} \quad \text{if } x \text{ only}$ $\mu(y) = e^{\int \frac{N_x - M_y}{M} \, dy} \quad \text{if } y \text{ only}$ |
| <b>Homogeneous</b><br>$\rightarrow$ separable in $v, x$ | $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$ $y = vx, \quad \frac{dy}{dx} = v + x \frac{dv}{dx}$                                                             |
| <b>Bernoulli</b><br>$\rightarrow$ linear in $w$         | $y' + P(x)y = Q(x)y^n, \quad w = y^{1-n}$ $\frac{dw}{dx} + (1-n)P(x)w = (1-n)Q(x)$                                                                          |

Linear Higher-Order ODEs

Zill 3.1–3.6

|                                                        |                                                                                                                      |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>General solution</b>                                | $y = y_c + y_p$ <p><math>y_c</math> solves the homogeneous eqn; <math>y_p</math> is any one particular solution.</p> |
| <b>Reduction of order</b><br>2nd sol. from known $y_1$ | $y'' + P(x)y' + Q(x)y = 0$ $y_2 = y_1(x) \int \frac{e^{-\int P \, dx}}{[y_1(x)]^2} \, dx$                            |

|                                                              |                                                                                                                                                                      |                                                            |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
|                                                              | <b>Const. coeff.</b><br>$ay'' + by' + cy = 0$                                                                                                                        | <b>Cauchy–Euler</b> , $x > 0$<br>$ax^2y'' + bxy' + cy = 0$ |
|                                                              | $y = e^{mx}$ :<br>$am^2 + bm + c = 0$                                                                                                                                | $y = x^m$ :<br>$am(m-1) + bm + c = 0$                      |
| <b>Distinct</b>                                              | $c_1e^{m_1x} + c_2e^{m_2x}$                                                                                                                                          | $c_1x^{m_1} + c_2x^{m_2}$                                  |
| <b>Repeated</b>                                              | $c_1e^{mx} + c_2xe^{mx}$                                                                                                                                             | $c_1x^m + c_2x^m \ln x$                                    |
| $\alpha \pm i\beta$                                          | $e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$                                                                                                                  | $x^\alpha[c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x)]$  |
| <b><math>n</math>th order, repeated</b><br>root of mult. $k$ | real $m$ : $e^{mx}, xe^{mx}, \dots, x^{k-1}e^{mx}$<br>pair $\alpha \pm i\beta$ : $x^je^{\alpha x} \cos \beta x, x^je^{\alpha x} \sin \beta x$<br>$j = 0, \dots, k-1$ |                                                            |

| $g(x)$                                                                        | trial $y_p$                       |
|-------------------------------------------------------------------------------|-----------------------------------|
| constant                                                                      | $A$                               |
| poly., degree $n$                                                             | $A_nx^n + \dots + A_1x + A_0$     |
| $e^{ax}$                                                                      | $Ae^{ax}$                         |
| $\cos \beta x$ or $\sin \beta x$                                              | $A \cos \beta x + B \sin \beta x$ |
| product of these                                                              | product of the forms              |
| A term of the guess already in $y_c$ ? Multiply the guess by $x$ (or $x^2$ ). |                                   |

|                                                           |                                                                                                                                                                               |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Variation of parameters</b><br>$y'' + Py' + Qy = f(x)$ | $y_c = c_1y_1 + c_2y_2, \quad W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$ $u_1' = \frac{-y_2f(x)}{W}, \quad u_2' = \frac{y_1f(x)}{W}$ $y_p = u_1y_1 + u_2y_2$ |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Laplace Transforms

Zill 4.1–4.5

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) dt$$

$t \geq 0$ ; converges for large  $s$  if  $f$  is piecewise continuous and of exponential order.

| $f(t)$             | $F(s)$                      | $f(t)$           | $F(s)$                            |
|--------------------|-----------------------------|------------------|-----------------------------------|
| 1                  | $\frac{1}{s}$               | $t^n$            | $\frac{n!}{s^{n+1}}$              |
| $e^{at}$           | $\frac{1}{s-a}$             | $e^{at} t^n$     | $\frac{n!}{(s-a)^{n+1}}$          |
| $\sin kt$          | $\frac{k}{s^2 + k^2}$       | $\cos kt$        | $\frac{s}{s^2 + k^2}$             |
| $\sinh kt$         | $\frac{k}{s^2 - k^2}$       | $\cosh kt$       | $\frac{s}{s^2 - k^2}$             |
| $t \sin kt$        | $\frac{2ks}{(s^2 + k^2)^2}$ | $t \cos kt$      | $\frac{s^2 - k^2}{(s^2 + k^2)^2}$ |
| $e^{at} \sin kt$   | $\frac{k}{(s-a)^2 + k^2}$   | $e^{at} \cos kt$ | $\frac{s-a}{(s-a)^2 + k^2}$       |
| $\mathcal{U}(t-a)$ | $\frac{e^{-as}}{s}$         | $\delta(t-a)$    | $e^{-as}$                         |

Valid for  $s > 0$ ;  $s > a$  when  $e^{at}$  appears;  $s > |k|$  for  $\sinh$ ,  $\cosh$ ;  $a \geq 0$  for  $\mathcal{U}$ ,  $\delta$ ;  $n = 1, 2, 3, \dots$

|                                                     |                                                                                                                                                                                      |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Linearity</b><br>also for $\mathcal{L}^{-1}$     | $\mathcal{L}\{af + bg\} = aF(s) + bG(s)$<br>$\mathcal{L}^{-1}\{aF + bG\} = a\mathcal{L}^{-1}\{F\} + b\mathcal{L}^{-1}\{G\}$<br>Inverse: partial fractions, then invert term by term. |
| <b>Derivatives</b><br>brings in<br>$y(0)$ , $y'(0)$ | $\mathcal{L}\{f'\} = sF(s) - f(0)$<br>$\mathcal{L}\{f''\} = s^2 F(s) - sf(0) - f'(0)$<br>$\mathcal{L}\{f^{(n)}\} = s^n F - s^{n-1} f(0) - \dots - f^{(n-1)}(0)$                      |
| <b>Integral</b>                                     | $\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{F(s)}{s}$                                                                                                                  |
| <b>1st shift</b><br>replace $s \rightarrow s-a$     | $\mathcal{L}\{e^{at} f(t)\} = F(s-a)$                                                                                                                                                |
| <b>Unit step</b>                                    | $\mathcal{U}(t-a) = \begin{cases} 0, & t < a \\ 1, & t \geq a \end{cases}$                                                                                                           |
| <b>2nd shift</b><br>delay by $a$                    | $\mathcal{L}\{f(t-a)\mathcal{U}(t-a)\} = e^{-as} F(s)$<br>$\mathcal{L}\{g(t)\mathcal{U}(t-a)\} = e^{-as} \mathcal{L}\{g(t+a)\}$                                                      |
| <b>Dirac delta</b><br>unit impulse                  | $\int_0^\infty f(t) \delta(t-a) dt = f(a)$<br>$\mathcal{L}\{\delta(t-a)\} = e^{-as}, \quad \mathcal{L}\{\delta(t)\} = 1$                                                             |
| <b>Convolution</b><br>product $F \cdot G$           | $(f * g)(t) = \int_0^t f(\tau) g(t-\tau) d\tau$<br>$\mathcal{L}\{f * g\} = F(s)G(s)$                                                                                                 |
| <b>Periodic</b><br>$f(t+T) = f(t)$                  | $\mathcal{L}\{f\} = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) dt$                                                                                                                    |

## SOLVING AN IVP BY LAPLACE

- 1 Transform both sides with  $\mathcal{L}\{y^{(n)}\} \rightarrow$  algebraic eqn in  $Y(s)$ .
- 2 Solve algebraically for  $Y(s)$ .
- 3 Partial fractions to match table entries; shift theorems for  $e^{\pm as}$  factors; convolution for a product  $F(s)G(s)$ .
- 4 Invert term by term:  $y(t) = \mathcal{L}^{-1}\{Y(s)\}$ .

## Vector Functions & Space Curves

Zill 9.1–9.3

**Vector function**  $\vec{r}(t) = f(t)\hat{i} + g(t)\hat{j} + h(t)\hat{k}$   
 tangent where  $\vec{r}' \neq \vec{0}$   $\vec{r}'(t) = f'(t)\hat{i} + g'(t)\hat{j} + h'(t)\hat{k}$

**Unit tangent**  
points along increasing  $t$   $\vec{T}(t) = \frac{\vec{r}'(t)}{\|\vec{r}'(t)\|}$

**Arc length**  

$$L = \int_a^b \|\vec{r}'(t)\| dt$$

$$= \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2} dt$$

**Arc-length fn. & speed**  
 $s(t) = \int_a^t \|\vec{r}'(u)\| du$

reparam. by  $s$  gives unit speed  $\frac{ds}{dt} = \|\vec{r}'(t)\| = v(t)$

**Curvature**  
cross form usually fastest  $\kappa = \frac{\|\vec{T}'(t)\|}{\|\vec{r}'(t)\|} = \frac{\|\vec{r}' \times \vec{r}''\|}{\|\vec{r}'\|^3}$

**Normal & binormal**  
right-handed frame  $\vec{T}, \vec{N}, \vec{B}$   
 $\vec{N}(t) = \frac{\vec{T}'(t)}{\|\vec{T}'(t)\|}, \quad \vec{B}(t) = \vec{T}(t) \times \vec{N}(t)$

**Frenet–Serret**  
 $\tau$  = torsion, twist out of osculating plane  
 $\frac{d\vec{T}}{ds} = \kappa\vec{N}, \quad \frac{d\vec{B}}{ds} = -\tau\vec{N}$   
 $\frac{d\vec{N}}{ds} = -\kappa\vec{T} + \tau\vec{B}$

**Velocity & accel.**  
 $a_T$  changes speed,  $a_N$  direction  
 $\vec{v} = \vec{r}', \quad \vec{a} = \vec{r}'' = a_T \vec{T} + a_N \vec{N}$   
 $a_T = v'(t), \quad a_N = \kappa[v(t)]^2$

## Fields: Gradient, Divergence, Curl

Zill 9.5–9.7

**Fields**  
 $\phi(x, y, z)$  scalar: temperature, pressure  
 $\vec{F} = P\hat{i} + Q\hat{j} + R\hat{k}$  vector: velocity, force

**Gradient**  
vector; max increase,  $\perp$  level surface  $f = c$   

$$\nabla f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

**Directional derivative**  
rate of change toward  $\vec{u}$   
 $D_{\vec{u}} f = \nabla f \cdot \vec{u}, \quad \|\vec{u}\| = 1$

## Divergence

scalar; outward  
flux/volume;  $> 0$   
source,  $< 0$  sink

$$\operatorname{div} \vec{F} = \nabla \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

## Curl

vector; local  
rotation;  $\vec{0}$   
everywhere  $\Rightarrow$   
irrotational

$$\operatorname{curl} \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P & Q & R \end{vmatrix}$$

## Identities

gradients  
irrotational; curls  
divergence-free

$$\nabla \times (\nabla f) = \vec{0}, \quad \nabla \cdot (\nabla \times \vec{F}) = 0$$

## Laplacian

harmonic if  
 $\nabla^2 f = 0$

$$\nabla^2 f = \nabla \cdot \nabla f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

## Line Integrals & Conservative Fields

Zill 9.8–9.9

### Scalar line integral

mass of a wire,  
avg. value

$$\int_C f(x, y, z) \, ds = \int_a^b f(\vec{r}(t)) \|\vec{r}'(t)\| \, dt$$

### Work

reversing  $C$   
negates it

$$W = \int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) \, dt$$
$$= \int_C P \, dx + Q \, dy + R \, dz$$

### Fundamental thm.

endpoints only;  
path-independent

$$\vec{F} = \nabla f \Rightarrow \int_C \vec{F} \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a))$$

### Conservative test

$\vec{F}$  cont. diff. on  
simply connected  
region

plane:  $\vec{F} = P\hat{i} + Q\hat{j}$  conservative

$$\iff \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

space: conservative  $\iff \nabla \times \vec{F} = \vec{0}$

### Potential function

same technique as  
exact DEs

$$f = \int P(x, y) \, dx + g(y)$$

match  $\frac{\partial f}{\partial y} = Q(x, y)$  to solve for  $g(y)$

### Closed loops

$$\vec{F} \text{ conservative} \iff \oint_C \vec{F} \cdot d\vec{r} = 0$$

for every closed curve  $C$

## Green's Theorem, Surface Integrals & Flux Zill 9.10–9.13

### Green's theorem

$C$  closed,  
counterclockwise,  
bounds  $D$ ;  $P, Q$   
cont. 1st partials

$$\oint_C P \, dx + Q \, dy = \iint_D \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

### Area via Green

$$A = \oint_C x \, dy = - \oint_C y \, dx = \frac{1}{2} \oint_C (x \, dy - y \, dx)$$

### Param. surface

$\vec{r}_u \times \vec{r}_v$  normal; its  
size scales area

$$\vec{r}(u, v) = x(u, v)\hat{i} + y(u, v)\hat{j} + z(u, v)\hat{k}$$

$$A(S) = \iint_D \|\vec{r}_u \times \vec{r}_v\| \, dA$$

### Surface integral

scalar field

$$\iint_S f \, dS = \iint_D f(\vec{r}(u, v)) \|\vec{r}_u \times \vec{r}_v\| \, dA$$

$$\text{for } z = g(x, y): \quad dS = \sqrt{g_x^2 + g_y^2 + 1} \, dA$$

### Flux

net flow of  $\vec{F}$   
through  $S$  along  $\hat{n}$

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_S \vec{F} \cdot \hat{n} \, dS$$
$$= \iint_D \vec{F}(\vec{r}(u, v)) \cdot (\vec{r}_u \times \vec{r}_v) \, dA$$

### Flux, graph

$z = g(x, y)$ ,  
upward  $\hat{n}$ ; flip sign  
for downward

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_D \left( -P \frac{\partial g}{\partial x} - Q \frac{\partial g}{\partial y} + R \right) dA$$

$D$  = projection of  $S$  onto the  $xy$ -plane.

## Divergence & Stokes' Theorems

Zill 9.14–9.16

### Divergence thm.

Gauss; closed  $S$   
bounds solid  $Q$ ,  
outward  $\hat{n}$

$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_Q \nabla \cdot \vec{F} \, dV$$

flux (surface)  $\rightarrow$  triple (volume) integral

### Stokes' thm.

$C$  bounds  $S$ ,  
oriented with  $\hat{n}$

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$$

circulation  $\rightarrow$  surface integral of curl

### Flat case

Green's = 2-D  
Stokes'

$$\hat{n} = \hat{k}, \quad S \subset xy\text{-plane} :$$
$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{k} \, dA$$

## COORDINATES & VOLUME ELEMENTS

### Cartesian

$$dA = dx \, dy, \quad dV = dx \, dy \, dz$$

### Polar

discs, annuli,  
sectors

$$x = r \cos \theta, \quad y = r \sin \theta, \quad dA = r \, dr \, d\theta$$

$$\iint_D f(x, y) \, dA$$
$$= \iint_D f(r \cos \theta, r \sin \theta) r \, dr \, d\theta$$

### Cylindrical

axis of symmetry

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z$$

$$dV = r \, dz \, dr \, d\theta$$

### Spherical

$\rho$  = dist. from  
origin;  $\phi \in [0, \pi]$   
from  $+z$ ;  $\theta$   
azimuthal

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

The Four Fundamental Theorems

boundary ↔ interior

All four: boundary integral = integral of a derivative over the region it bounds — the ordinary FTC, one dimension up each time.

|                          |                                                                                  |
|--------------------------|----------------------------------------------------------------------------------|
| Line integrals           | $\int_C \nabla f \cdot d\vec{r} = f(\text{end}) - f(\text{start})$               |
| $\vec{F} = \nabla f$     |                                                                                  |
| Green's                  | $\oint_C P \, dx + Q \, dy = \iint_D (Q_x - P_y) \, dA$                          |
| $C = \partial D$ , plane |                                                                                  |
| Stokes'                  | $\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\text{curl } \vec{F}) \cdot d\vec{S}$ |
| $C = \partial S$         |                                                                                  |
| Divergence               | $\iiint_S \vec{F} \cdot d\vec{S} = \iiint_Q (\text{div } \vec{F}) \, dV$         |
| $S = \partial Q$ , solid |                                                                                  |

WHICH THEOREM APPLIES?

| you want...                                                            | use        |
|------------------------------------------------------------------------|------------|
| $\int_C \vec{F} \cdot d\vec{r}$ on an open curve, $\vec{F} = \nabla f$ | Fund. thm. |
| $\oint_C P \, dx + Q \, dy$ , $C$ bounds plane region $D$              | Green's    |
| $\oint_C \vec{F} \cdot d\vec{r}$ , $C$ bounds a surface $S$ in space   | Stokes'    |
| flux $\iint_S \vec{F} \cdot d\vec{S}$ , $S$ closed around solid $Q$    | Divergence |